

Smart Dustbin

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Project Idea



Smart Automated Waste Segregation System

We propose a **Smart Dustbin** that utilizes IoT to automate waste management at the source. The system identifies, classifies, and physically segregates waste without human intervention.

Classification Categories

 Organic

 Paper

 Plastic

 Other



Detection

Ultrasonic sensors detect waste entry and trigger the system.



Capture & Inference

Camera module captures image → Raspberry Pi 4 runs [EfficientNetB0](#) model to predict category.



Sorting Action

Servo motors tilt the base plate to slide waste into the predicted partition.

Project Approach



PHASE 1: COMPLETED

Data & AI Model

- ✓ Dataset collection and cleaning (corrupt image removal)
- ✓ Model training using
- ✓ Split: 80% Train, 10% Validation, 10% Test



PHASE 2: COMPLETED

Edge Deployment

- ✓ Deployed model on **Raspberry Pi 4**
- ✓ Optimized for real-time inference
- ✓ Integration with Python environment



PHASE 3: COMPLETED

Physical Build

- Ultrasonic Sensors for waste detection
- Camera module integration for image capture
- Chassis construction



PHASE 4: COMPLETED

Sorting Mechanism

- Servomotor actuation logic
- Tilting plate mechanism
- 4 Categories: **Paper, Organic, Metal, Other**

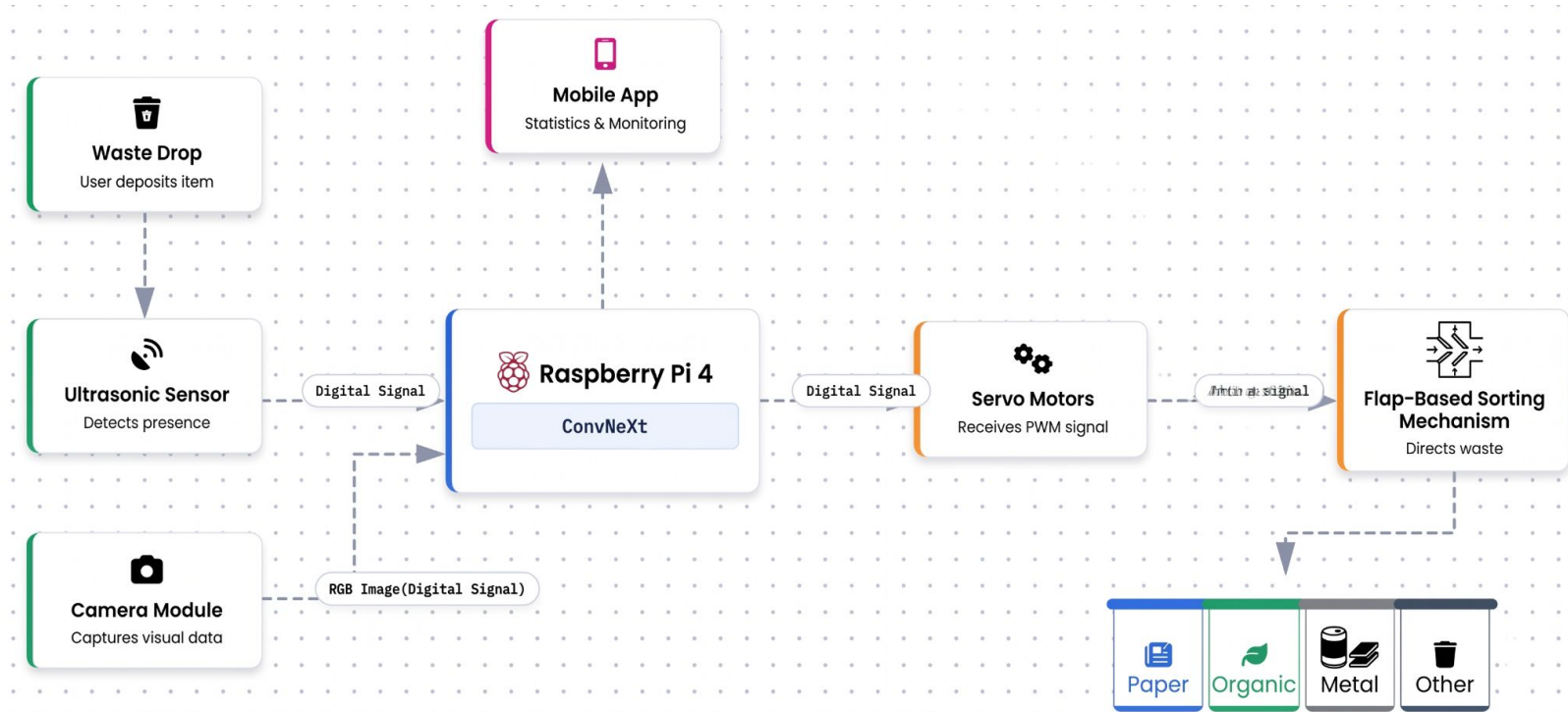
Bill of Materials (BOM)



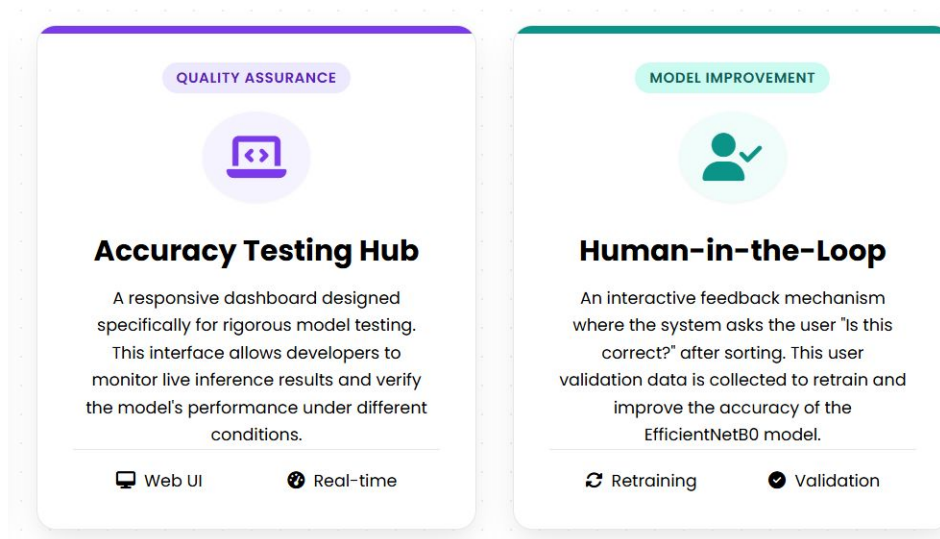
S.No.	Item	Qty *	Cost ^	Link	Description
1	Raspberry Pi 4 Model B	1	₹ 5,073.00	Raspberry Pi 4 Model B	Edge inference device (runs TFLite model)
2	Ultrasonic sensor	1	₹ 192.00	Ultrasonic sensor	HC-SR04 Sensor
3	Camera module	1	₹ 1,698.00	Camera module	Raspberry Pi V2 Camera Module
4	Servo motors	4	₹ 250.00	Servo motors	SG90 Servo Motors
5	Corrugated plastic	6 sheets	₹ 275.00	Corrugated plastic	Mechanical Tilting Plate
6	Power Supply	1	₹ 699.00	Power supply	12V 5200 mAh Li-ion Battery
7	Buck converter	1	₹ 44.00	Buck converter	
8	Wires & connectors	1 set	NA	NA	Assorted Cables And Jumping Wires
9	Prototyping components	1 set	NA	NA	Breadboard, Resistors etc,

^ Items with "NA" in the "Cost" column, along with some of the other items were issued from the ECE Labs, IIIT Delhi.

Block Diagram



Possible Enhancements



Dataset: <https://drive.google.com/drive/folders/18WbJjeRjZWBh0ciaezF-8BIWj1eWP6mj?usp=sharing>

Augmented Dataset: <https://drive.google.com/file/d/1EHqlwZiGdSAMzCCM3LbEV-xLImh1w-OZ/view?usp=sharing>

Iteration 1: Initial Model & Problem Identification



1. OBJECTIVE

🧠 Build AI waste classifier (Paper, Plastic, Organic, Other)

2. APPROACH

📦 MobileNetV2 Transfer Learning

🖼️ Basic Augmentation (Flip, Rotation, Zoom)

3. KEY PROBLEMS IDENTIFIED

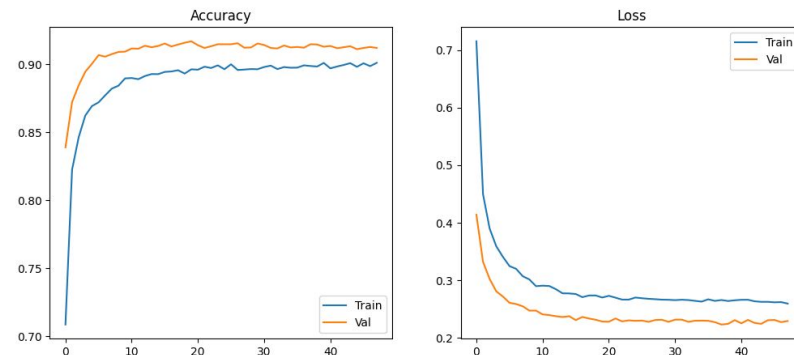
❗ Misclassification as "Paper"

⚠️ Poor real-world generalization

4. LEARNING

💡 Accuracy alone is misleading; Model bias exists

PERFORMANCE & DATASET INSIGHTS



ACCURACY & LOSS CURVES

~93%

VAL ACCURACY



CLASSES : ORGANIC | PAPER | PLASTIC

Iteration 2: From High Accuracy to Real-World Failure



MODEL SUCCESS

MobileNet → **EfficientNetB0**

97%

Test Accuracy

- Real-time Flask + Mobile Cam
- **~90% Confidence**



MAJOR FAILURE

50-60%

Accuracy Drop

REAL-WORLD DEPLOYMENT FAILURE

- Pi NoIR Camera used
- Pink tint (No IR filter)

Model failed in real-world conditions



FIX & INSIGHT

Class Redesign:

Old: Paper, Plastic, Organic, Other
New: **Paper, Metal, Organic, Other**

- Added CutMix Augmentations
- Identified dataset mismatch



Pi NoIR Raw Capture

KEY INSIGHT: Dataset must match real hardware input (**Pi NoIR camera**) for reliable performance.

Iteration 3: Model Optimization & System Integration



AI Model Evolution

Data & Architecture Refinement

- Shifted to grayscale images to reduce color dependency (NoIR camera) for the **EfficientnetB0** model.
- Focused on shape and edge-based learning.
- Switched to **ConvNeXt** architecture with SupCon.
- Final Classes: Paper, Metal, Organic, Other.

~90%

Real-world Pi Data

98%+

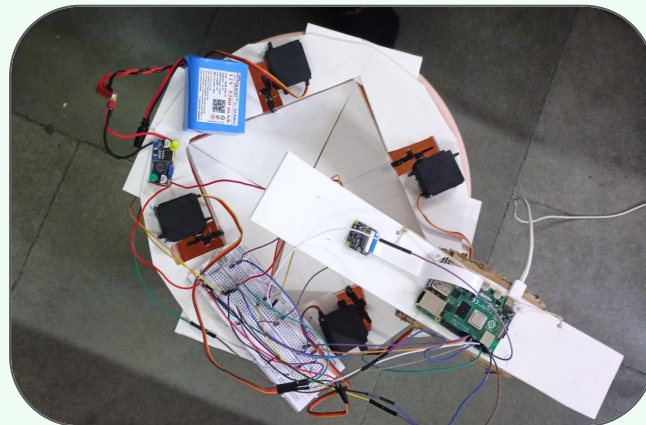
Final Accuracy

STATUS: FULLY INTEGRATED

Image capture → classification → motor actuation working seamlessly in real-world conditions.



Hardware Development



Engineering Solutions

- **Wiring:** Switched from taped and loose wires to breadboard.
- **Calibration:** Fixed unpredictable motor startup by calibrating servos to a common starting angle.
- **Stability:** Improved structural stability by focusing on certain balance points.

Challenges, Learnings & System Evolution

Key Challenges

- ❗ **Domain Gap:** Training vs Pi NoIR input
- 🎮 **Color Distortion:** Pink tint issues
- 📉 **Model Bias:** Early paper class favor
- ⚡ **Hardware Instability:** Misalignment

Mechanical Evolution

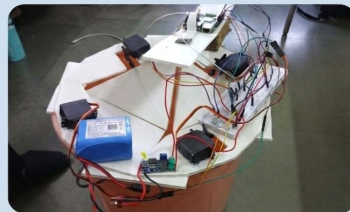
Tilt Plate → **Flap-Based Sorting Mechanism**



- Redesigned from circular tilt to square structure
- Each flap corresponds to one waste category
- Improved control, precision, and scalability

Key Learnings

- ✅ **Real-World Testing** is essential
- 📷 Match hardware conditions
- 📊 Accuracy ≠ performance
- 🔧 Mechanical reliability is key



AI accuracy + Mechanical design = Reliable real-world system

Real World Drawbacks



Relative Deployment Cost

Scaling across Indian cities remains a major investment, despite being cheaper than market alternatives.



ML Classification Limits

Variability, contamination, and overlapping features challenge standalone model accuracy.



Multi-Sensor Fusion Needs

Adding metal detection and IR/laser sensors is required, increasing system complexity.



Single-Item Processing

System constrained to one item at a time, differing from typical simultaneous disposal behavior.



Real-World Variability

Dust, lighting shifts, and wet waste reduce performance compared to controlled lab tests.



User Behavior Dependency

Relies on proper item-by-item placement, which is difficult to enforce in public spaces.



KEY INSIGHT: A hybrid system combining ML with sensor-based detection and user-centric design is essential for real-world deployment.

Work Distribution



- **Ashmit Verma (2024133)** : Trained and optimized multiple deep learning models (MobileNet → EfficientNet) for waste classification, converted the best-performing model to TensorFlow Lite, and deployed it on Raspberry Pi. Managed system setup and contributed to hardware implementation. Created and curated the dataset used for training on the Raspberry Pi, along with a separate “pink dataset” for testing and evaluating model performance. Also developed and presented Midsem and Endsem evaluation PPTs.
- **Anurag Basistha (2024092)** : Built datasets (mainly Kaggle) via keyword scraping and filtering; implemented ingestion with CLIP, YOLO, and Qwen2 VLM, with SHA deduplication and t-SNE validation. Trained ConvNeXtV2 (Atto → Nano) achieving 98%+ accuracy under 2M params and 90%+ per-class confidence; attempted MAE pretraining (VRAM-limited) and fine-tuned fcmae.in1k (SupCon → cross-entropy). Developed an Android demo (mock data) with class selection, map-based bin visualization, and time-filtered multi-bin analytics; planned Raspberry Pi integration for real inference.
- **Chinmaya Sharma (2024163)** : Worked on multiple ideas during the ideation phase. Developed with the physical design of the dustbin and implemented it. Came up with the circuitry and hardware components. Implemented the hardware portion of the project. Prepared the Midsem and Endsem evaluation PPTs. Made the schematic and the PCB layout for the project.

Thank You



We sincerely express our heartfelt gratitude to Professor Alok Mittal for his constant guidance, encouragement, and support throughout this project. We also extend our thanks to our Teaching Assistants, Krishaang Phogat and Arnav Gupta, for their valuable insights and continuous assistance.



Ashmit Verma



Anurag Basistha



Chinmaya Sharma

— From your beloved students —

Ashmit Verma | Anurag Basistha | Chinmaya Sharma

Thank you for your time and attention.